

3. Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

Aim

To use NumPy arrays and operations to find the average sale price of houses having more than four bedrooms in a neighborhood.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array containing the number of bedrooms and sale price of each house.
4. Use a condition to select houses having **more than 4 bedrooms**.
5. Extract the sale prices of those houses.
6. Calculate the average using `np.mean()`.
7. Display the average sale price.
8. Stop the program.

Python Program :

```
import numpy as np
```

```
# House data
```

```
# Column 1 = Bedrooms
```

```
# Column 2 = Sale Price
```

```
houses = np.array([
```

```
[3, 250000],
[5, 450000],
[4, 350000],
[6, 550000],
[7, 650000],
[2, 200000]
])

# Select houses with more than 4 bedrooms
selected_houses = houses[houses[:, 0] > 4]

# Get sale prices
prices = selected_houses[:, 1]

# Calculate average sale price
average_price = np.mean(prices)

print("Houses with more than 4 bedrooms:")
print(selected_houses)

print("\nAverage sale price:", average_price)
```

Output :

```
===== RESTART: C:/Users/T.Jaya Krishna Reddy/AppData/Local/Programs/Python/Python314/exp 1.py =====  
Houses with more than 4 bedrooms:  
[[ 5 450000]  
 [ 6 550000]  
 [ 7 650000]]  
  
Average sale price: 550000.0  
>
```